NAME

CUTEST_csj_threaded − CUTEst tool to evaluate constraint Jacobian values in a sparse format.

SYNOPSIS

CALL CUTEST_csj_threaded(status, n, X, nnzj, lj, J_val, J_var, J_con, thread)

For real rather than double precision arguments, instead

CALL CUTEST_csj_threaded_s(...)

and for quadruple precision arguments, when available,

CALL CUTEST_csj_threaded_q(...)

DESCRIPTION

The CUTEST_csj_threaded subroutine evaluates the values of the Jacobian J(x) of the constraint functions c(x) of the problem decoded from a SIF file by the script *sifdecoder* at the point X in the constrained minimization case. The Jacobian is stored in a sparse format.

The problem under consideration is to minimize or maximize an objective function $f(x)$ over all $x \in R^n$ subject to general equations $c_i(x) = 0$, $(i \in 1, \ldots, m_E)$, general inequalities $c_i^l \leq c_i(x) \leq c_i^u$, $(i \in m_E + 1, \ldots, m)$, and simple bounds $x^l \leq x \leq x^u$. The objective function is group-partially separable and all constraint functions are partially separable.

ARGUMENTS

The arguments of CUTEST_csj_threaded are as follows

status [out] - integer
: the outputr status: 0 for a succesful call, 1 for an array allocation/deallocation error, 2 for an array bound error, 3 for an evaluation error, 4 for an out-of-range thread,

n [in] - integer
: the number of variables for the problem,

X [in] - real/double precision
: an array which gives the current estimate of the solution of the problem,

nnzj [out] - integer
: the number of nonzeros in J_val,

lj [in] - integer
: the actual declared dimensions of J_val, J_var and J_con,

J_val [out] - real/double precision
: an array which gives the values of the nonzeros of the general constraint functions evaluated at X. The i-th entry of J_val gives the value of the derivative with respect to variable J_var(i) of constraint function J_con(i),

J_var [out] - integer
: an array whose i-th component is the index of the variable with respect to which J_val(i) is the derivative,

J_con [out] - integer
an array whose i-th component is the index of the constraint function of which J_val(i) is the derivative,

thread [in] - integer
thread chosen for the evaluation; threads are numbered from 1 to the value threads set when calling CUTEST_csetup_threaded.

AUTHORS

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SEE ALSO

CUTEst: a Constrained and Unconstrained Testing Environment with safe threads,
N.I.M. Gould, D. Orban and Ph.L. Toint,
Computational Optimization and Applications **60**:3, pp.545-557, 2014.

CUTEr (and SifDec): A Constrained and Unconstrained Testing Environment, revisited,
N.I.M. Gould, D. Orban and Ph.L. Toint,
ACM TOMS, **29**:4, pp.373-394, 2003.

CUTE: Constrained and Unconstrained Testing Environment,
I. Bongartz, A.R. Conn, N.I.M. Gould and Ph.L. Toint,
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cutest_ccfsg_threaded(3M), sifdecoder(1)